

Comparative Analysis of Breast Cancer Risk Factors among African-American Women and White Women

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The authors assessed risk factor profiles among 1,505 African-American and 1,809 White women in the 1993–2001 Carolina Breast Cancer Study. Multiple logistic regression models for case-control data were used to estimate odds ratios for several factors. Racial differences were observed in the prevalence of many breast cancer risk factors among both younger (aged 20–49 years) and older (aged 50–74 years) women. For older women, the magnitude and direction of associations were generally similar for African-American and White women, but important racial differences were observed among younger women. In particular, multiparity was associated with increased risk of breast cancer among younger African-American women (for three or four pregnancies: adjusted odds ratio (OR) = 1.5, 95% confidence interval (CI): 0.9, 2.6; for five or more pregnancies: OR = 1.4, 95% CI: 0.6, 3.1) but not among younger White women (for three or four pregnancies: OR = 0.7, 95% CI: 0.4, 1.2; for five or more pregnancies: OR = 0.8, 95% CI: 0.2, 3.0). The relations with age at first full-term pregnancy and nulliparity also varied by race. Case-only analyses before and after further adjustment for tumor stage and hormone receptor status revealed little effect on results. Hence, racial variations in both prevalences of and risks associated with particular factors may contribute to the higher incidence of breast cancer among younger African-American women.

African Americans; breast neoplasms; case-control studies; risk factors; women

Abbreviations: CI, confidence interval; OR, odds ratio.

Few published epidemiologic studies have examined whether racial or ethnic differences exist in the prevalence of risk factors for breast cancer or the magnitude of their associations with the disease. Of case-control studies that have analyzed breast cancer risk factors among African-American women (1–6), only two were population based and examined African-American women and White women in the same study (3, 6). Both of these studies examined risk factors among younger women (aged 20–54 years) and, thus, had limited ability to elucidate racial differences by age or menopausal status.

Distinguishing breast cancers by age at onset has important implications for cancer incidence and etiology. The most recent Surveillance, Epidemiology, and End Results

Program statistics for 1996–2000 found the age-adjusted incidence of breast cancer to be higher among White than African-American women (142.0 per 100,000 vs. 120.8 per 100,000) (7). The same pattern was seen in a comparison of White and African-American women aged 50 years or older (398.6 vs. 322.1 per 100,000), but the trend was reversed among women aged less than 40 years. In that group, the incidence rate for each 5-year age group was higher among African-American women (7). This disparity indicates that African-American women are experiencing excess rates of breast cancer at younger ages. Evaluation of age distributions in case series has shown that African-American breast cancer patients are more likely to present at a younger age (8–11) and that cases of breast cancer among young women

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overall are more aggressive and show a poorer prognosis and response to treatment than cases among older women (8, 11–15). These factors contribute to a mortality rate for breast cancer among younger African-American women that is twice that of younger White women (7).

To evaluate potential differences in risk factors for breast cancer, we compared risk factor profiles between African-American women and White women, stratified at age 50 years, in the Carolina Breast Cancer Study. To highlight differences in risk factors that may contribute to excess incidence among younger African-American women, we first present adjusted odds ratios for reproductive and lifestyle factors. Because breast tumor characteristics are known to differ between African-American and White women (16), we then analyzed the pattern of results from case-only analyses before and after further adjustment for stage and hormone receptor status.

MATERIALS AND METHODS

Subjects

The Carolina Breast Cancer Study is a population-based, case-control study of breast cancer in African-American and White women residing in a 24-county area in central and eastern North Carolina (17). Both cases and controls were sampled using a modification of randomized recruitment (18). All women between 20 and 74 years of age who were diagnosed with a first, invasive breast cancer between May 1, 1993, and December 31, 2000, were eligible as cases and were identified using the North Carolina Central Cancer Registry's Rapid Case Ascertainment System (19). Sampling probabilities ensured approximately equal samples in the four age-race groups: younger (age 20–49 years) African-American women, older (age 50–74 years) African-American women, younger White women, and older White women. We excluded women of other races, who constituted less than 2 percent of the study population. Of the 2,397 eligible and locatable cases, 172 (7 percent) were denied contact with us by their physicians, and an additional 352 (15 percent) declined to participate. Thus, information from 1,873 cases was available for analysis, for an overall cooperation rate of 78 percent. Age- and race-specific cooperation rates were 84 percent, 80 percent, 76 percent, and 72 percent for younger White and African-American cases and older White and African-American cases, respectively.

Controls were drawn from North Carolina Division of Motor Vehicles lists for women aged 20–64 years and US Health Care Financing Administration lists for women aged 65–74 years. Sampling probabilities for controls ensured approximate frequency matching to cases by race and 5-year age groups. Of the 2,444 eligible and locatable women, 727 (30 percent) declined to participate. Thus, information from 1,717 controls was available for analysis, for an overall cooperation rate of 70 percent. Cooperation rates for the four age-race groups were 76 percent, 71 percent, 72 percent, and 63 percent for younger White and African-American controls and older White and African-American controls, respectively.

After exclusion of 276 women who agreed only to a brief telephone survey and had incomplete information on many risk factors of interest, the final data set consisted of 1,505 (45 percent) African-American women (787 cases and 718 controls) and 1,809 (55 percent) White women (991 cases and 818 controls).

Data collection

The data were obtained during in-person interviews conducted by female registered nurses. Through a questionnaire, the nurse-interviewer elicited information on demographics and potential breast cancer risk factors, including first-degree family history of breast and ovarian cancer, menstrual and reproductive history, and sociodemographic and lifestyle characteristics. The nurses drew a blood sample and measured weight, height, and waist and hip circumferences at the time of interview. The median time from diagnosis to interview for the cases was 3 months (range: 1–19 months); 80 percent were interviewed within 5 months of diagnosis. For the controls, the median time from selection to interview was 2 months (range: 0–26 months); 80 percent were interviewed within 5 months of selection.

The American Joint Committee on Cancer stage was abstracted from medical records, where available, or determined from information on tumor size, lymph node involvement, and distant metastasis (16). Estrogen receptor and progesterone receptor status was obtained from medical records for 80 percent of the cases. For the remaining cases, estrogen receptor and progesterone receptor status was determined with paraffin-embedded tumor tissues at the University of North Carolina laboratory (11 percent), or receptor status was missing (9 percent) (19, 20).

Statistical methods

Statistical analyses were performed separately by race, and they were further stratified by age or menopausal status. Women were categorized as postmenopausal if they reported natural menopause or bilateral oophorectomy or if they were more than 55 years of age and reported hysterectomy. Women who reported still having menstrual cycles or who had at least one remaining ovary and were aged less than 42 years were classified as premenopausal. Women who had a hysterectomy without bilateral oophorectomy and were aged 42–55 years were considered perimenopausal.

The frequency distributions of risk factors in the study sample were adjusted using age-specific sampling weights to estimate prevalences in the underlying population. Comparisons by race were evaluated using the chi-square statistic. Both the prevalence estimates and chi-square tests were generated using SUDAAN version 8.0.0 software (Research Triangle Institute, Research Triangle Park, North Carolina). Odds ratios with 95 percent confidence intervals were calculated using logistic regression models that examined the association between breast cancer status and risk factors after adjustment for other relevant covariates including age (continuous), age at menarche, parity, age at first full-term pregnancy, miscarriage, breastfeeding, induced abortion, oral contraceptive use, and hormone replacement therapy.

TABLE 1. Prevalence* of breast cancer risk factors among controls, by race and age, Carolina Breast Cancer Study, 1993–2001

	20–49 years				<i>p</i> value	50–74 years				<i>p</i> value
	African American		White			African American		White		
	No.	%	No.	%		No.	%	No.	%	
Age group (years)										
20–34	39	37	35	49						
35–44	172	42	189	34						
45–49	153	21	180	17						
50–54						80	30	92	31	
55–64						148	40	145	37	
65–74						126	30	177	32	
Age at menarche (years)										
≤11	91	25	72	16		76	22	61	16	
12–13	176	49	244	51		170	49	232	56	
≥14	97	26	87	33	0.36	105	28	117	27	0.13
Missing			1	0		3	1	4	1	
Menopausal status†										
Pre-/perimenopausal	311	92	352	94		46	17	66	22	
Postmenopausal, ≤44 years	37	6	39	5		133	35	122	28	
Postmenopausal, 45–49 years	16	2	13	1	0.31	83	23	117	27	
Postmenopausal, ≥50 years						92	25	109	23	0.04
Age (years) at first full-term pregnancy										
≤24	243	60	190	37		273	78	271	65	
25–29	39	11	98	20		33	10	81	20	
≥30	34	8	55	11		14	4	30	7	
Nulliparous	47	21	61	32	<0.01	32	8	32	8	<0.01
Missing	1	0				2	0			
Parity										
0	47	21	61	32		32	8	32	8	
1–2	192	49	269	53		108	32	193	48	
3–4	102	26	68	14		92	28	155	37	
≥5	23	5	6	1	<0.01	122	32	34	7	<0.01
Breastfed										
Never	278	79	231	63		184	55	246	62	
Ever	86	21	173	37	<0.01	170	45	168	38	0.04
Oral contraceptive use (months)										
<3	74	20	50	10		228	61	205	42	
3–59	135	37	147	44		80	25	119	33	
60–120	97	23	133	32		34	11	54	15	
>120	56	20	71	14	<0.03	11	3	31	9	<0.01
Missing	2	0	3	0		1	0	5	1	

Table continues

Body mass index, waist/hip ratio, history of breast cancer in a first-degree relative, education, alcohol consumption, and smoking were also included as covariates in addition to a term for the sampling fraction. Analyses also were adjusted for years since last full-term pregnancy; these results are not presented because odds ratios were equivalent within plus or

minus 0.1. Logistic analyses were performed using SAS PROC GENMOD software (SAS Institute, Inc., Cary, North Carolina), which permits the use of an offset term to take the sampling design into account.

Tests for interaction between each covariate and race were conducted by performing the likelihood ratio test using data

TABLE 1. Continued

	20–49 years					<i>p</i> value	50–74 years					<i>p</i> value
	African American		White		African American		White					
	No.	%	No.	%	No.		%	No.	%			
Family history of breast cancer in first-degree relative												
No	319	86	354	77	0.21	293	83	344	83	0.90		
Yes	34	8	47	23		44	12	52	13			
Missing	11	6	3	0		17	5	18	4			
Body mass index (kg/m ²)‡												
<25	57	14	188	53	<0.01	41	11	168	41	<0.01		
≥25	294	82	208	46		303	86	241	58			
Missing	13	4	8	1		10	3	5	1			
Waist/hip ratio§												
<0.81	166	53	292	74	0.01	112	32	245	60	<0.01		
≥0.81	190	45	108	25		237	67	165	39			
Missing	8	2	4	1		5	1	4	1			
Education (years)												
≤12	178	52	120	24	<0.01	250	68	165	37	<0.01		
13–15	102	24	121	41		60	18	145	36			
≥16	84	24	163	35		43	13	104	27			
Missing						1	0					
Hormone replacement												
Never	313	92	334	91	0.78	236	66	176	41	<0.01		
Ever	51	8	70	9		118	34	238	59			
Alcohol												
Never	116	39	68	16	<0.01	160	44	152	35	0.02		
Ever	247	61	336	84		194	56	262	65			
Missing	1	0										
Smoking												
Never	214	67	196	40	<0.01	211	58	202	49	<0.05		
Former	61	11	121	40		91	26	135	32			
Current	89	22	87	20		52	16	77	19			
Abortion												
Induced												
Never	301	82	334	81	0.77	334	94	396	95	0.41		
Ever	63	18	70	19		20	6	18	5			
Spontaneous												
Never	266	73	311	80	0.33	229	66	291	71	0.13		
Ever	98	27	93	20		125	34	123	29			

* Frequencies adjusted for sampling weights. Totals may not add to 100 because of rounding.

† Women were categorized as postmenopausal if they reported natural menopause or bilateral oophorectomy, or if they were more than 55 years of age and had a hysterectomy. Women who reported still having menstrual cycles or who had at least one remaining ovary and were less than 42 years of age were classified as premenopausal. Women who had a hysterectomy without bilateral oophorectomy and who were 42–55 years of age were considered perimenopausal. The variable also captures age at menopause of postmenopausal women.

‡ Body mass index of less than 25 was defined as normal weight by the National Heart, Lung, and Blood Institute (32).

§ Median value among controls = 0.81.

with both races combined. A model containing only the main effects of race, a variable of interest, and other covariates was compared with a model containing the main effects of

race, the variable of interest, the relevant interaction term, and other covariates. Tests were conducted for interaction among all women, younger women, and older women. None

TABLE 2. Race-specific, adjusted odds ratios for breast cancer and reproductive and lifestyle risk factors among women aged 20–49 years, Carolina Breast Cancer Study, 1993–2001

	African American (n = 753)						White (n = 972)					
	Cases (n = 389)		Controls (n = 364)				Cases (n = 568)		Controls (n = 404)			
	No.	%	No.	%	Adjusted odds ratio*	95% confidence interval	No.	%	No.	%	Adjusted odds ratio	95% confidence interval
Age group (years)												
20–34	56	14	39	11			61	11	35	9		
35–44	182	47	172	47			263	46	189	47		
45–49	151	39	153	42			244	43	180	45		
Age at menarche (years)												
≥14	79	20	97	27	1.0		105	18	87	22	1.0	
12–13	210	54	176	48	1.6	0.9, 2.4	333	59	244	60	1.2	0.8, 1.7
≤11	100	26	91	25	1.6	1.0, 2.4	129	23	72	18	1.7	1.1, 2.6
Missing							1	0	1	0		
Menopausal status†												
Pre-/perimenopausal	343	88	311	85	1.0		511	90	352	87	1.0	
Postmenopausal, ≤44 years	29	7	37	10	0.7	0.4, 1.3	27	5	39	10	0.4	0.2, 0.8
Postmenopausal, 45–49 years	17	4	16	4	0.8	0.4, 1.7	30	5	13	3	1.4	0.7, 2.9
Age at first full-term pregnancy (years)‡												
≤24	263	67	243	67	1.0		240	42	190	47	1.0	
25–29	44	11	39	11	1.2	0.7, 2.0	130	23	98	24	1.1	0.7, 1.6
≥30	28	7	34	9	0.7	0.4, 1.4	88	15	55	14	1.3	0.9, 2.1
Nulliparous	54	14	47	13	0.8	0.5, 1.3	110	19	61	15	1.4	0.9, 2.2
Missing				1	0							
Parity												
0	54	14	47	13	1.0		110	19	61	15	1.0	
1–2	192	49	192	53	1.1	0.7, 1.8	363	64	269	67	0.8	0.5, 1.2
3–4	120	31	102	28	1.5	0.9, 2.6	88	15	68	17	0.7	0.4, 1.2
≥5	23	6	23	6	1.4	0.6, 3.1	7	1	6	1	0.8	0.2, 3.0
Breastfed												
Never	324	83	278	76	1.0		341	60	231	57	1.0	
Ever	65	17	86	24	0.6	0.4, 0.8	227	40	173	43	1.0	0.7, 1.4
Oral contraceptive use (months)												
<3	71	18	74	20	1.0		68	12	50	12	1.0	
3–59	141	36	135	37	1.1	0.7, 1.8	214	38	147	36	1.2	0.8, 1.8
60–120	111	29	97	27	1.1	0.7, 1.7	183	32	133	33	1.0	0.7, 1.7
>120	66	17	56	15	1.3	0.8, 2.2	101	18	71	18	1.1	0.7, 1.9
Missing			2	1			2	0	3	1		
Family history of breast cancer in first-degree relative												
No	319	82	319	88	1.0		460	81	354	88	1.0	
Yes	61	16	34	9	2.2	1.3, 3.5	92	16	47	12	1.7	1.1, 2.5
Missing	9	2	11	3			16	3	3	1		

Table continues

of the interactions by race reached statistical significance at $p < 0.05$; however, a few comparisons yielded likelihood ratio tests with $p < 0.20$ (noted in text). Because of the limited statistical power for tests of interaction, we also chose to highlight differences between risks for African-American women and White women of 40 percent or greater

in magnitude of the odds ratio or odds ratios that went in opposite directions. These criteria, though admittedly arbitrary, were selected to avoid making too much of fairly small variations in odds ratios and to focus attention on variations in odds ratios by race that were potentially meaningful in our data. We thought such results warranted further examination

TABLE 2. Continued

	African American (n = 753)						White (n = 972)					
	Cases (n = 389)			Controls (n = 364)			Cases (n = 568)			Controls (n = 404)		
	No.	%	No.	%	Adjusted odds ratio	95% confidence interval	No.	%	No.	%	Adjusted odds ratio	95% confidence interval
Body mass index (kg/m ²)§												
<25	77	20	57	16	1.0		308	54	188	47	1.0	
≥25	305	78	294	81	0.7	0.5, 1.1	252	44	208	51	0.7	0.5, 0.9
Missing	7	2	13	4			8	1	8	2		
Waist/hip ratio¶												
<0.81	156	40	166	46	1.0		393	69	292	72	1.0	
≥0.81	229	59	190	52	1.4	1.0, 1.9	166	29	108	27	1.4	1.0, 2.0
Missing	4	1	8	2			9	2	4	1		
Education (years)												
≤12	192	49	178	49	1.0		162	28	120	30	1.0	
13–15	109	28	102	28	1.1	0.7, 1.6	164	29	121	30	0.9	0.7, 1.4
≥16	88	23	84	23	1.1	0.7, 1.8	242	43	167	40	1.0	0.7, 1.4
Missing												
Hormone replacement												
Never	342	88	313	86	1.0		475	84	334	83	1.0	
Ever	47	12	51	14	1.0	0.6, 1.6	3	16	70	17	1.1	0.7, 1.7
Alcohol												
Never	127	33	116	32	1.0		108	19	68	17	1.0	
Ever	262	67	247	68	0.9	0.6, 1.3	460	81	336	83	0.8	0.6, 1.2
Missing			1	0								
Smoking												
Never	218	56	214	59	1.0		305	54	196	49	1.0	
Former	90	23	61	17	1.8	1.2, 2.7	169	30	121	30	0.9	0.7, 1.3
Current	81	21	89	24	1.0	0.6, 1.4	94	17	87	22	0.7	0.5, 1.0
Abortion												
Induced												
Never	305	78	301	83	1.0		465	82	334	83	1.0	
Ever	84	22	63	17	1.2	0.8, 1.8	103	18	70	17	1.0	0.7, 1.4
Spontaneous												
Never	293	75	266	73	1.0		441	78	311	77	1.0	
Ever	93	25	98	27	1.0	0.7, 1.4	127	22	93	23	1.1	0.8, 1.5

* These odds ratios were obtained from a logistic regression model adjusted for an offset term, age (continuous), and all other variables in the table except age at first full-term pregnancy.

† Women were categorized as postmenopausal if they reported natural menopause or bilateral oophorectomy, or if they were more than 55 years of age and had a hysterectomy. Women who reported still having menstrual cycles or who had at least one remaining ovary and were less than 42 years of age were classified as premenopausal. Women who had a hysterectomy without bilateral oophorectomy and who were 42–55 years of age were considered perimenopausal. The variable also captures age at menopause of postmenopausal women.

‡ Age at first full-term pregnancy was entered separately into a model containing all the other variables except parity.

§ Body mass index of less than 25 was defined as normal weight by the National Heart, Lung, and Blood Institute (32). The logistic model for body mass index did not include waist/hip ratio.

¶ Median value among controls = 0.81.

in the discussion in relation to possible consistency with other available findings.

To adjust for racial differences in breast cancer characteristics (i.e., stage at diagnosis and hormone receptor status), we conducted a series of case-only analyses using logistic regression models. Odds ratios and 95 percent

confidence intervals were derived from case-case comparisons to highlight the presence of heterogeneity between African-American and White women with breast cancer after adjustment for stage and estrogen receptor and progesterone receptor status (i.e., odds ratios deviating from 1.0 suggest racial differences). Although the case-

TABLE 3. Race-specific, adjusted odds ratios for breast cancer and reproductive and lifestyle risk factors among women aged 50–74 years, Carolina Breast Cancer Study, 1993–2001

	African American (n = 752)						White (n = 837)					
	Cases (n = 398)		Controls (n = 354)				Cases (n = 423)		Controls (n = 414)			
	No.	%	No.	%	Adjusted odds ratio*	95% confidence interval	No.	%	No.	%	Adjusted odds ratio	95% confidence interval
Age group (years)												
50–54	87	22	80	23			84	20	92	22		
55–64	164	41	148	42			169	40	145	35		
65–74	147	37	126	36			170	40	177	43		
Age at menarche (years)												
≥14	105	26	105	30	1.0		105	25	117	28	1.0	
12–13	193	48	170	48	1.2	0.8, 1.8	245	58	232	56	1.2	0.9, 1.7
≤11	99	25	76	21	1.3	0.8, 2.0	73	17	61	15	1.5	0.9, 2.3
Missing	1	0	3	1					4	1		
Menopausal status†												
Pre-/perimenopausal	38	10	46	13	1.0		51	12	66	16	1.0	
Postmenopausal, ≤44 years	129	32	133	38	1.4	0.7, 2.5	110	26	122	29	1.5	0.8, 2.6
Postmenopausal, 45–49 years	93	23	83	23	1.6	0.8, 3.0	123	29	117	28	1.7	1.0, 2.9
Postmenopausal, ≥50 years	138	35	92	26	2.2	1.2, 4.0	139	33	109	26	2.2	1.3, 3.8
Age at first full-term pregnancy (years)‡												
≤24	280	70	273	77	1.0		250	59	271	65	1.0	
25–29	37	9	33	9	1.0	0.6, 1.8	85	20	81	20	1.2	0.8, 1.8
≥30	18	5	14	4	1.0	0.5, 2.3	39	9	30	7	1.4	0.8, 2.5
Nulliparous	57	14	32	9	2.0	1.1, 3.6	48	11	32	8	1.5	0.9, 2.6
Missing	6	2	2	1			1	0				
Parity												
0	57	14	32	9	1.0		48	11	32	8	1.0	
1–2	109	27	108	31	0.5	0.3, 0.9	194	46	193	47	0.7	0.4, 1.2
3–4	108	27	92	26	0.5	0.3, 1.0	152	36	155	37	0.7	0.4, 1.2
≥5	124	31	122	34	0.5	0.3, 0.9	29	7	34	8	0.8	0.4, 1.6
Breastfed												
Never	236	59	184	52	1.0		261	62	246	59	1.0	
Ever	162	41	170	48	0.8	0.6, 1.2	162	38	168	41	0.9	0.7, 1.3
Oral contraceptive use (months)												
<3	259	65	228	64	1.0		214	51	205	50	1.0	
3–59	71	18	80	23	0.8	0.5, 1.3	101	24	119	29	0.8	0.6, 1.2
60–120	45	11	34	10	1.4	0.8, 2.5	67	16	54	13	1.1	0.7, 1.8
>120	23	6	11	3	2.1	0.9, 4.8	37	9	31	7	1.2	0.7, 2.2
Missing							4	1	5	1		
Family history of breast cancer in first-degree relative												
No	325	82	293	83	1.0		334	79	344	83	1.0	
Yes	57	14	44	12	1.2	0.8, 1.9	78	18	52	13	1.5	1.0, 2.2
Missing	16	4	17	5			11	3	18	4		

Table continues

case odds ratio can be used as a measure of heterogeneity of odds ratios (21), its magnitude reflects risk factor differences in both background prevalence and disease-associated risk, and the absence of noncase comparisons

limits etiologic inferences. However, comparison of unadjusted and adjusted case-only odds ratios can provide an indication as to whether the tumor characteristics are potentially important to racial differences in risk factors.

TABLE 3. Continued

	African American (n = 752)						White (n = 837)					
	Cases (n = 398)		Controls (n = 354)		Adjusted odds ratio	95% confidence interval	Cases (n = 423)		Controls (n = 414)		Adjusted odds ratio	95% confidence interval
	No.	%	No.	%			No.	%	No.	%		
Body mass index (kg/m ²)§												
<25	55	14	41	12	1.0		172	41	168	41	1.0	
≥25	323	81	303	86	0.8	0.5, 1.5	247	58	241	58	0.9	0.7, 1.3
Missing	20	5	100	3			4	1	5	1		
Waist/hip ratio¶												
<0.81	104	26	112	32	1.0		232	55	245	59	1.0	
≥0.81	285	72	237	67	1.5	1.0, 3.0	187	44	165	40	1.3	0.9, 1.7
Missing	9	2	5	1			4	1	1	1		
Education (years)												
≤12	283	71	250	71	1.0		183	43	165	40	1.0	
13–15	65	16	60	17	1.0	0.7, 1.6	124	29	145	35	0.8	0.6, 1.1
≥16	50	13	43	12	0.9	0.5, 1.5	116	27	104	25	1.0	0.7, 1.4
Missing	1	0										
Hormone replacement												
Never	290	73	236	67	1.0		185	44	176	43	1.0	
Ever	108	27	118	33	0.6	0.4, 0.8	238	56	238	57	0.9	0.7, 1.2
Alcohol												
Never	166	42	160	45	1.0		155	37	152	37	1.0	
Ever	232	58	194	55	1.1	0.7, 1.5	267	63	262	63	1.0	0.7, 1.5
Missing							1	0				
Smoking												
Never	218	55	211	60	1.0		187	44	202	49	1.0	
Former	126	32	91	26	1.2	0.8, 1.8	173	41	135	33	1.4	1.0, 1.9
Current	54	14	52	15	0.9	0.5, 1.4	63	15	77	19	0.8	0.5, 1.2
Abortion												
Induced												
Never	380	95	334	94	1.0		403	95	396	96	1.0	
Ever	18	5	20	6	1.0	0.5, 2.0	20	5	18	4	1.0	0.5, 2.1
Spontaneous												
Never	251	63	229	65	1.0		324	77	291	70	1.0	
Ever	147	37	125	35	1.1	0.8, 1.6	99	23	123	30	0.7	0.5, 1.0

* These odds ratios were obtained from a logistic regression model adjusted for an offset term, age (continuous), and all the other variables in the table except age at first full-term pregnancy.

† Women were categorized as postmenopausal if they reported natural menopause or bilateral oophorectomy, or if they were more than 55 years of age and had a hysterectomy. Women who reported still having menstrual cycles or who had at least one remaining ovary and were less than 42 years of age were classified as premenopausal. Women who had a hysterectomy without bilateral oophorectomy and who were 42–55 years of age were considered perimenopausal. The variable also captures age at menopause of postmenopausal women.

‡ Age at first full-term pregnancy was entered separately into a model containing all the other variables except parity.

§ Body mass index of less than 25 was defined as normal weight by the National Heart, Lung, and Blood Institute (32). The logistic model for body mass index did not include waist/hip ratio.

¶ Median value among controls = 0.81.

RESULTS

The prevalences of various risk factors among controls for younger (aged 20–49 years) and older (aged 50–74 years) African-American women and White women are shown in table 1. Results stratified by menopausal status were very similar to the age-stratified results presented here. Among

younger women, numerous statistically significant differences in the distribution of risk factors were seen by race, with African-American women younger at first full-term pregnancy and having more births, less likelihood of breast-feeding, less but longer use of oral contraceptives, larger body size, higher waist/hip ratio, lower levels of education, less

alcohol consumption, and less smoking. No significant differences were seen in the distribution of age at menarche, menopausal status or age at menopause, first-degree family history, hormone replacement therapy usage, induced abortion, or miscarriage, although some differences were observed for prevalence of age at menarche and family history. Among younger parous women, the distributions for time since last full-term pregnancy were identical for African-American and White women: 8 percent, less than or equal to 3 years; 20 percent, 4–10 years; and 72 percent, 11 or more years. In general, the same similarities and differences were observed among older women. Exceptions to the pattern seen for younger women were that older African-American women underwent menopause at earlier ages (on average) than did White women and were more likely to have breastfed and less likely to have used hormone replacement therapy than their White peers. Among older women, all except three reported their age since last full-term pregnancy of greater than or equal to 11 years, and these three were between 4 and 10 years.

Comparison of prevalence estimates between younger and older women revealed differences in reproductive and other behaviors. Regardless of race, younger women were more educated, delayed childbearing, and had fewer children. Use of oral contraceptives, induced abortion, and alcohol use were much more common among younger women. Among African Americans, breastfeeding and miscarriage were lower in the younger group. Younger Whites were more likely to have ever smoked than were older Whites, and older African Americans were more likely to have quit smoking than were younger African Americans. In addition, older women of both races had more abdominal obesity (as measured by waist/hip ratio) and were more likely to have used hormone replacement therapy.

In table 2, associations between breast cancer and various reproductive and lifestyle characteristics are shown for younger women by race; differences in odds ratios by race were observed for most reproductive characteristics and also former smoking. Younger African-American women with average age at menarche (12–13 years) were at increased risk of breast cancer, as were their White counterparts (but not to the same degree). For younger White women, the risk increased with later age at first pregnancy, and nulliparous women were at greater risk than were parous women. In contrast, among African-American women, later age at first pregnancy did not increase risk, and nulliparous African Americans were at slightly reduced risk versus those who had a first pregnancy before age 25 years and those with three or more children. African-American women who breastfed were at reduced risk of breast cancer, but White women were not ($p_{\text{interaction}} < 0.12$). An interaction with race was also suggested for age at menopause, with odds ratios farther from 1.0 at both younger (aged ≤ 44 years) and later (aged 45–49 years) ages among younger White women than younger African-American women ($p_{\text{interaction}} < 0.19$). In addition, former African-American smokers were at elevated risk, but former White smokers were not ($p_{\text{interaction}} < 0.06$). Patterns and magnitudes of risk were similar by race for family history of breast cancer, body size, waist/hip ratio, education, oral contraceptive and hormone replacement

therapy use, alcohol consumption, and induced and spontaneous abortion.

Associations among older women were more similar by race for reproductive factors, but there was some evidence that they varied for nulliparity, oral contraceptive use, hormone replacement therapy use, and spontaneous abortion (table 3). As was true for younger women, older African-American women contrasted with their White counterparts by not having increased risk from late age at first full-term pregnancy. Among nulliparous older women, however, both races were at higher risk than parous women, and the odds ratio for nulliparous African-American women was 50 percent higher than the odds ratio for their White counterparts. The odds ratio for extended oral contraceptive use was 90 percent higher among older African-American women than White women. In contrast, hormone replacement therapy use was significantly inversely associated for African-American women (odds ratio (OR) = 0.6, 95 percent confidence interval (CI): 0.4, 0.8) but had no association for White women ($p_{\text{interaction}} < 0.07$). For spontaneous abortion, the interaction by race was suggestive ($p < 0.14$), with the odds ratio for White women reporting one or more events equaling 0.7, whereas for African-American women there was essentially no association. Patterns and magnitudes of risk were similar for other risk factors, and 95 percent confidence intervals generally included 1.0, except for the twofold increase in risk with later age at menopause (≥ 50 years) in both races.

Among both younger and older women with breast cancer, African Americans were more likely than Whites to be diagnosed with advanced stage disease, and their tumors were more likely to be negative for hormone receptors (table 4). Case-only analyses were used to determine whether these differences in breast cancer characteristics might explain, at least in part, racial differences in associations of risk factors with disease. However, adjustment for both stage and hormone receptor status had minimal impact on case-case odds ratios, including odds ratios for all of the reproductive and lifestyle risk factors found to have associations with breast cancer that differed by race in case-control analyses (data not shown). Of the 15 risk factors assessed (comprising 26 odds ratios for various levels of exposure), only two odds ratios adjusted for estrogen receptor and progesterone receptor status among younger women and one odds ratio adjusted for stage among older women varied by 15 percent from their respective values prior to these adjustments. Adjustment for both stage and hormone receptor status made no further difference, none of these odds ratios differed by more than 20 percent from the comparison value, and all resulted in attenuated differences in risk factors between African-American and White cases.

DISCUSSION

A crossover in breast cancer incidence at 45–49 years of age has been observed for African-American women and White women, with African Americans having the higher rate at younger ages and lower incidence after age 50 years relative to White women (7, 22). In a recent review of the literature,

TABLE 4. Case-only ratios for tumor characteristics and race, stratified by age, Carolina Breast Cancer Study, 1993–2001

	20–49 years						50–74 years					
	African American		White		Adjusted odds ratio*	95% confidence interval	African American		White		Adjusted odds ratio	95% confidence interval
	No.	%	No.	%			No.	%	No.	%		
Stage at diagnosis												
I	105	27	219	39	1.0		147	37	230	54	1.0	
II	194	50	248	44	1.7	1.2, 2.3	172	43	141	33	1.9	1.4, 2.5
III, IV	71	18	68	12	2.4	1.6, 3.7	50	13	26	6	2.8	1.7, 4.8
Missing	19	5	33	6			29	7	26	6		
ER†/PR‡ status												
ER+†/PR+†	107	27	255	45	1.0		162	41	225	53	1.0	
ER–†/PR–†	169	43	157	28	3.1	2.2, 4.3	114	29	69	16	2.2	1.5, 3.2
Other‡	56	14	76	13	1.9	1.3, 3.0	53	13	72	17	1.0	0.7, 1.5
Missing	60	15	80	14			69	17	57	13		

* The odds ratios were obtained from a logistic regression model adjusted for age (continuous), age at menarche, a combined variable of menopausal status and age at menopause, parity, breastfeeding, oral contraceptive use, body mass index, waist/hip ratio, education, hormone replacement, alcohol, smoking, induced and spontaneous abortion, and an offset term.

† ER, estrogen receptor; PR, progesterone receptor; ER+, estrogen receptor positive; PR+, progesterone receptor positive; ER–, estrogen receptor negative; PR–, progesterone receptor negative.

‡ “Other” refers to ER+/PR– and ER–/PR+.

Pathak et al. (23) concluded that this crossover should be expected rather than being considered an anomaly, given the age-specific distributions for age at first full-term pregnancy and parity in the two racial groups. The expectation of crossover is based on the theory that a transient increase in risk follows pregnancy, and a subsequent long-term protective effect follows, generally for 10 years postpregnancy, because of differentiation of the breast ducts. Because African-American women have more children at younger ages, the risk of breast cancer for younger African-American women would be expected to be higher than that of White women of similar age (23). Our results expand upon the hypothesis of Pathak et al. by confirming racial differences in the distributions of several risk factors (also reported recently by Bernstein et al. (24)) as well as uncovering possible age-specific differences between African-American and White women in the magnitude of risk conferred by certain risk factors. We propose that the combined effect of a greater prevalence of some risk factors and relatively higher odds for breast cancer from certain risk factors may be responsible for a greater number of breast cancer cases among young African-American women.

Several reproductive risk factors were among those with apparent variations in prevalence and patterns of breast cancer risk between African Americans and Whites. Older age at first full-term pregnancy was associated with a modest increased risk among White women, with nulliparous women being at highest risk, as is traditionally reported. This was evident for both younger and older women among Whites; however, the pattern did not hold for African-American women. Among younger African Americans, no associations with older age at first full-term pregnancy or nulliparity were observed. Among older African-American women, increasing age at first full-term pregnancy showed no association with breast cancer, but nulliparous women

were at twice the risk of parous women. Palmer et al. (25) also found no elevation of risk due to late age at first birth in older African-American women. Brinton et al. (6) obtained similar results, reporting that nulliparous younger African-American women were at reduced risk of breast cancer relative to multiparous women. In addition, Laing et al. (4) reported no increased breast cancer risk for nulliparous African-American women compared with multiparous African-American women. Hence, it appears that neither age at first full-term pregnancy nor nulliparity helps to explain the observed racial pattern of breast cancer incidence.

Multiparity is often reported to reduce breast cancer risk in comparison with women having no children (26). We observed this relation among both younger and older White women, although there was no evidence of a dose-response trend. Again, however, relations among African-American women tended to vary by age. While a similar inverse relation was observed for older African Americans, younger African-American women with three or more children had an odds ratio 40–50 percent higher (not statistically significant, however) than the odds ratio for their nulliparous peers. They also were almost twice as likely to have families of this size as younger White women. These results replicate the findings of Palmer et al. (25), who documented that parity is associated with increased risk in younger African-American women but with decreased risk among older women. An age-dependent effect of increasing parity on breast cancer risk has been reported previously (27–29). In one study, Bruzzi et al. (27) provided evidence for a positive association between breast cancer and increasing parity among women younger than age 40 years with two or more children, but (as with our data for African Americans) the trend was not significant, while they reported an inverse relation for older women. These authors also reported a transient

increase in breast cancer risk following full-term pregnancy; a relative risk of 2.66 (95 percent CI: 1.31, 5.39) was obtained for women having delivered a child within 3 years versus women whose last child was born 10 or more years earlier. If younger African-American women have children at intervals of 3 years or less, the transient risk period could be prolonged. In addition, if having multiple children actually increases risk of breast cancer for young African-American women, the higher prevalence of this factor among African Americans could serve to elevate risk for breast cancer in this population, consistent with the higher incidence of breast cancer reported for younger African-American women.

If, as a consequence of higher parity, young African-American women were more likely to have a recent pregnancy (≤ 10 years previously) than were young White women, the relevant exposure may be time since last full-term pregnancy. These age-specific differences in odds ratios between African-American and White women might be expected if age since last full-term pregnancy is the relevant exposure and if younger African-American women were more likely to have a recent pregnancy. However, among both younger and older parous women in our study, there were no racial differences in distribution of time since last full-term pregnancy. Additionally, when the variable years since last full-term pregnancy was substituted for age at first full-term pregnancy, we saw the expected trend of an increased risk followed by a progressive reduction in odds ratios for younger White women, but no association was seen for younger African-American women (data not shown). Moreover, the odds ratios for parity were virtually unchanged after adjustment (data not shown). We had insufficient variability in time since last full-term pregnancy to conduct similar analyses for older women. Nevertheless, these results provide additional evidence for variation by race in relations between breast cancer and characteristics of reproductive history among younger women.

Previous results from the Carolina Breast Cancer Study on the effect of breastfeeding on breast cancer risk showed a 30 percent reduction for parous women who had ever breastfed relative to parous women who had not (30). Our race-specific analyses showed younger African-American women to be at greater reduced risk (OR = 0.6, 95 percent CI: 0.4, 0.8) than their White peers (OR = 1.0, 95 percent CI: 0.7, 1.4). The inverse association also was stronger among younger African-American women than their older counterparts (OR = 0.6 and OR = 0.8 for younger and older, respectively). We noted that, among older women, the prevalence of breastfeeding was higher among African-American than White women. In contrast, only 20 percent of younger African-American women, almost half the proportion of younger White women, had breastfed. Thus, many younger African-American women may not be benefiting from the potential protective effects of lactation.

Selection bias may have influenced our results, as response rates varied across age, race, and case status (31). Women who declined an in-person interview were asked to complete a brief telephone survey on basic breast cancer risk factors. Both cases and controls who responded only to the telephone survey were older, had an earlier age at first full-

term pregnancy, and had less education, oral contraceptive use, and hormone replacement therapy use than women who participated fully. However, the differences were limited and in the same direction for cases and controls, minimizing the concern about selection bias (31). Moreover, comparisons by race for prevalence of risk factors when based only on women with complete data provide conservative estimates of differences. A further refinement took into account differences in disease characteristics between African Americans and Whites in supplementary case-only analyses. Adjustment for stage at disease diagnosis and hormone receptor status of the breast cancer failed to reduce racial differences in risk factor profiles among cases, suggesting that any differences in etiologic pathways are unlikely to be explained by stage at diagnosis or estrogen and progesterone receptor status alone.

Even with the relatively large numbers of cases and controls in this study, the sample sizes were modest for many comparisons when analyses were stratified by race and age; hence, confidence intervals were broad and firm conclusions not possible. Although the findings we highlighted did not represent statistically significant differences by race, we thought it was important to focus on point estimates to discern whether any patterns may emerge. This was particularly of interest because we had roughly comparable numbers of African-American and White women from the same geographic area. With the increasing availability of studies that have included reasonable numbers of African-American women (1–6, 25), a meta-analysis of results may provide more meaningful estimates of risk factor profiles by race.

In conclusion, published reports on the etiology of breast cancer among African-American women are sparse and often conclude that their risk factors are similar to those for White women. Our results, however, show racial differences in the prevalence of most risk factors. Furthermore, the associations between breast cancer and some of these factors appear to vary in magnitude and direction by race and age. Most of the variations in risk factors occurred among women under 50 years of age, and the observed differences cannot be explained by either stage at diagnosis or hormone receptor status. Our results therefore demonstrate the importance of assessing modification by age when discerning risk estimates, and they support the hypothesis that racial variations in risk combined with racial differences in prevalence for particular risk factors may contribute to the higher incidence of breast cancer among younger African-American women.

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